Mock exam 1

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Does the speed with which the car is moving affect the distance needed to stop?

You work on the development of an algorithm that can identify vehicles that move at potentially dangerous speeds using data from cameras. Your part of the project is to develop a suitable statistical model. To train the algorithm, your colleagues measured the speed of various cars (in miles) and the distance they pass before they stop (in feet).

The original data can be found in the `cars` dataset (`data(cars)`).

Questions

- Load the dataset and convert columns to the metric form – from feet to meters and miles per hour to kilometers per hour. 1 meter is equal to 3.28 feet and 1 mile is equal to 1609.344 meters. (3 points.)
- Build the appropriate regression model. Check the assumptions for the use of this model. (5 points.)
- Check for any abnormal values. Justify your decision (3 points).
- If the assumptions are satisfied, discuss the model. How much variance is explained? Is the correlation statistically significant? What is the relationship between the speed and the distance to stop? (5 points.)
- Build a plot that shows all the data points and depicts the model on it. (6 points.)
- The cameras equipped with your algorithm captured 5 cars. Their speed is 35 and 68 km/h. What is the predicted distance needed to stop? Use R to make these predictions. (3 points.)

Marking

Load the dataset and modify it according to the requirements – 3 points

- Load the data – 1 point
- Convert the columns correctly – 2 points

```
data("cars")
Cars = cars

Cars$dist = Cars$dist / 3.28
Cars$speed = Cars$speed * 1.609
```

Build a suitable regression model and check its assumptions – 5 points

- Formulate the regression model – 2 points
- Clearly state and test the assumptions – 3 points

The assumption for regressions:

- Independence of observations – can assume at once
- The relationship between the variables is linear
- The model residuals do not change with the fitted value
- The model residuals are normally distributed

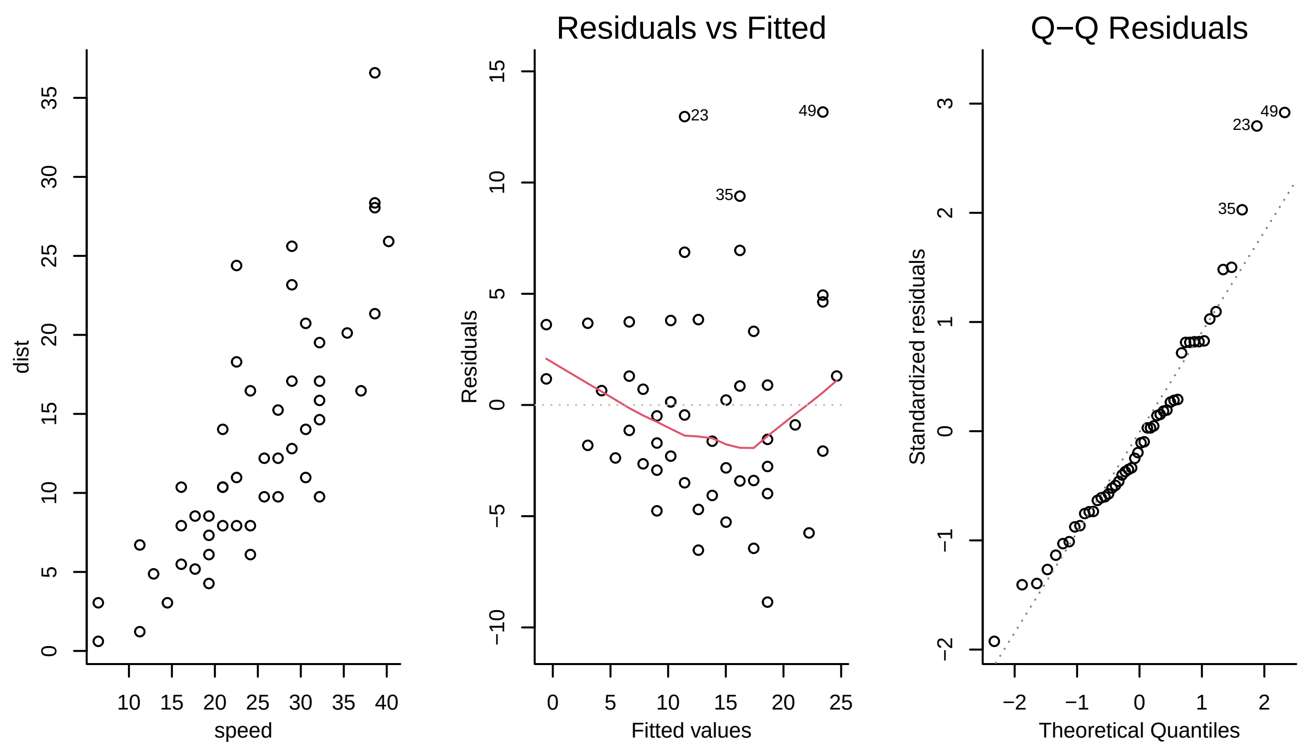


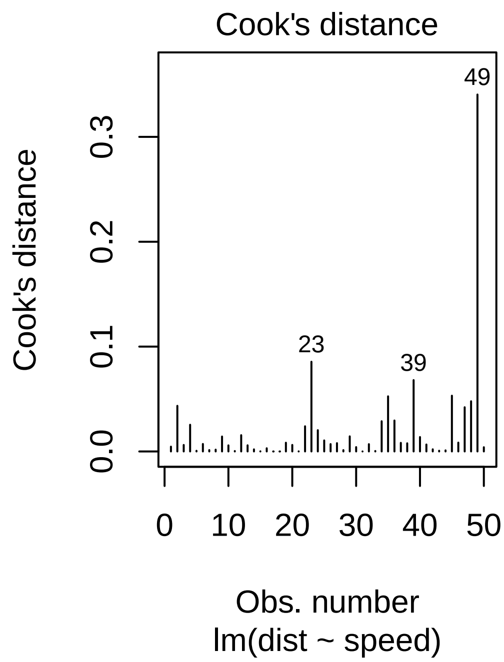
Figure 1: Analyzing the model.

- The relationship between the variables seems to be quite linear according to the scatter plot.
- The model residuals are not strictly homoscedastic.
- The model residuals have some values that do not strictly correspond to the expected values.

Overall, the model is usable. However, some values seem to be out of place. For instance, there are really few measurements from the cars moving at a lower speed. It skews the model.

Check for outliers – 3 points

This task is for reasoning. Some values can be considered outliers indeed. However, the key point is that there are no indications that these values are wrong or obtained due to an error. Thus, if you wish to exclude them, explain why you did it.



For example, I am not going to exclude anything. Removal of observation 49 would make sense, but there are no clear reasons for this measurement to be abnormal. This value makes the model more "pessimistic". In my opinion, it would be better to be safe than sorry.

Discuss the model – 5 points

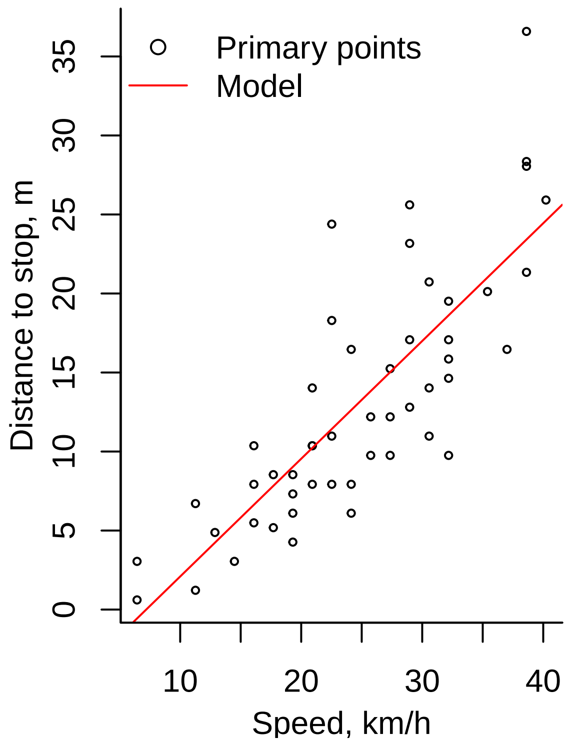
```
Call:
lm(formula = dist ~ speed, data = Cars)

Residuals:
    Min      1Q  Median      3Q     Max
-8.8625 -2.9041 -0.6926  2.8094 13.1711

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -5.35948    2.06050  -2.601   0.0123 *
speed        0.74512    0.07873   9.464 1.49e-12 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.689 on 48 degrees of freedom
Multiple R-squared:  0.6511,    Adjusted R-squared:  0.6438
F-statistic: 89.57 on 1 and 48 DF,  p-value: 1.49e-12
```

The model is statistically significant. The variables correlate at $r \approx 0.8$. The model explains around 65% of the total variance. Thus, the model is quite good, but there is still quite a lot of unexplained variance. It would be better to account for more factors.

Make a plot – 6 points

As always, the marking is done as follows:

- 1 point for making any plot
- 2 points for making a plot that has either poor design or is uninformative.
- 3 points for a clear scatterplot
- +1-2 points for designing it well and including the legend
- +1 point for adding the model to it



Making predictions – 3 points

This task is straightforward. It would not harm to notice that the model requires quite extensive extrapolation. It is not good.

```
(Intercept) (Intercept)
   20.71987    45.30898
```

It is possible to use the R-specific function (`predict(...)`). But it will not change the mark.

Originally created by Dmytro Shytikov in 2025.